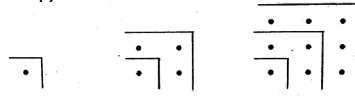


AMTI (NMTC) - 2004
GAUSS CONTEST - PRIMARY LEVEL

1. Look at the following dot diagram



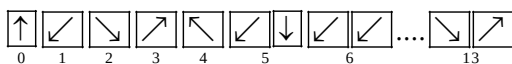
This pattern continues. The value of $1 + 3 + 5 + \dots$ up to 100 terms is the number of dots shown in the

- (A) 100th diagram and the number of dots present in it is 1000
 (B) 1000th diagram and the number of dots present in it is 10,000
 (C) 100th diagram and the number of dots present in it is 10,000
 (D) 1000th diagram and the number of dots present in it is 1000
2. Look at the rows of numbers shown below :

1st row :	1	$1 = \frac{1 \times 2}{2}$
2nd row :	2 3	$3 = \frac{2 \times 3}{2}$
3rd row :	4 5 6	$6 = \frac{3 \times 4}{2}$
4th row :	7 8 9 10	$10 = \frac{4 \times 5}{2}$ and so on

The first number in the 50th row is

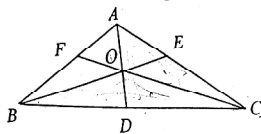
- (A) 1275 (B) 1224 (C) 1276 (D) 1226
3. In the sequence 1, 22, 333, 10101010101010101010, 11111111111111111111,, the sum of the digits in the 200th term is
 (A) 200 (B) 400 (C) 600 (D) 40000
4. How many two digit numbers greater than 10 are there, which are divisible by 2 and 5 but not by 4 or 25 ?
 (A) 3 (B) 12 (C) 5 (D) 2
5. The number of 3 digit even numbers that can be written using the digits 0, 3, 6 without repetition is
 (A) 6 (B) 3 (C) 4 (D) 2
6. In the sequence of numbers 1, 2, 11, 22, 111, 222, the sum of the digits in the 999th term is
 (A) 999 (B) 1998 (C) 500 (D) 1000
7. When 1000 single digit non-zero numbers are added, the units place is 5. The maximum carry over in this case is
 (A) 495 (B) 895 (C) 899 (D) 995
8. You can write the number 1 using 5 and 7 and by addition and subtraction as $5 + 5 + 5 - 7 - 7 = 1$ (or) $7 + 7 + 7 - 5 - 5 - 5 - 5 = 1$ and so on. But using 3 fives and 2 sevens is the best ways as we are using totally 5 numbers, whereas in the second example, we use 7 numbers. Using the above method, if 1 is written using the digits 2's and 5's only, the minimum number of times 2's and 5's are used is
 (A) three 2's and one 5 (B) three 5's and seven 2's
 (C) thirteen 2's and five 5's (D) two 2's and one 5
9. 4ab5 is a four digit number divisible by 55 where a, b are unknown digits. Then $b - a$ is
 (A) 1 (B) 4 (C) 5 (D) 0
10. In the Fee - Vee land, the numbers are written as follows :



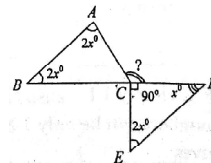
then ↙ ↘ represents

- (A) 7 (B) 9 (C) 14 (D) 21

11. The sum of the reciprocals of all the divisors of 6 is
 (A) 1 (B) 2 (C) less than 2 (D) greater than 2
12. In the adjoining figure, the number of triangles formed is



- (A) 6 (B) 7 (C) 10 (D) 16
13. In the figure shown B, C, D lie on the same line. $m\angle ECD = 90^\circ$.
 $m[2\angle CDE] = m\angle CDE$
 $= m\angle BAC$
 $= m\angle ABC$



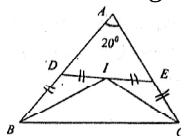
The value of $m\angle ACD$ is

- (A) 120° (B) 150° (C) 210° (D) 240°
14. If distinct numbers are replaced for distinct letters in the following subtraction,

$$\begin{array}{r} \text{FOUR} \\ - \text{ONE} \\ \hline \text{TWO} \end{array}$$

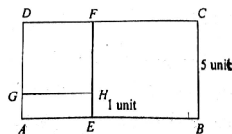
then the values of F and T are given by

- (A) $F = 1, T = 9$
 (B) $F = 1, T = 8$
 (C) $F = 1$ and T is any single digit other than 1
 (D) F and T cannot be determined
15. $\triangle ABC$ is an isosceles triangle with $m\angle A = 20^\circ$ and $AB = AC$. D and E are points on AB and AC such that $AD = AE$. I is the midpoint of the segment DE.



If $BD = ID$, then the angles of $\triangle IBC$ are

- (A) $110^\circ, 35^\circ, 35^\circ$ (B) $100^\circ, 40^\circ, 40^\circ$
 (C) $80^\circ, 50^\circ, 50^\circ$ (D) $90^\circ, 45^\circ, 45^\circ$
16. In the figure given BCFE, DF EA are squares, $BC = 5$ units, $HE = 1$ unit, the length and breadth of the rectangle ABCD are



- (A) 8 units and 5 units (B) 5 units and 10 units
 (C) 5 units and 7 units (D) 9 units and 5 units
17. If $!5 = 4 + 6 - 5$, $!12 = 11 + 13 - 12$ and $!23 = 22 + 24 - 23$, then what is the value of $!40 + !41 + !42 + !43 + !44 + \dots + !49 + !50$?
 (A) 505 (B) 495 (C) 455 (D) 465
18. The number of pairs of two digit square numbers, the sum or difference of which are also square numbers is
 (A) 0 (B) 1 (C) 2 (D) 3
19. $(a - 1)^2 + (b - 2)^2 + (c - 3)^2 + (d - 4)^2 = 0$. Then $a \times b \times c \times d + 1$ is
 (A) 0^2 (B) 10^2
 (C) 5^2 (D) $1^2 + 2^2 + 3^2 + 4^2 + 1$
20. The adjacent table defines the operation ; e.g. $b * c = a$, $d * a = a$ etc.

*	a	b	c	d
a	b	c	d	a
b	c	d	a	b
c	d	a	b	c
d	a	b	c	d

If $b * x = d$ then $(x * c) * a$ is

- (A) a (B) b (C) c (D) d

